INT 330 -- MANAGING CLOUD SOLUTIONS

CA 2: PROJECT

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Designing a Simple AWS Cloud Architecture.

The goal of this project is to design a simple and efficient architecture using Amazon Web Services (AWS) to demonstrate the key components of cloud infrastructure. The architecture will focus on scalability, reliability, and cost-effectiveness, utilizing fundamental/advance AWS services. The project will provide hands-on experience in deploying cloud solutions and understanding AWS best practices…...
This project will serve as an introduction to building cloud architectures, providing a foundation for more complex AWS solutions in the future…...

Basic AWS Services to be used

- Amazon S3 – **As Storage**
- Amazon EC2 **– As Compute service**
- AWS Identity and Access Management (IAM) (with restrictions) **- As Security service**
- AWS VPC – **As Network service**
- AWS RDS – **As Database service**

Advanced AWS Services:

- AWS Lambda
- Amazon CloudFront
- Amazon Elastic Load Balancer
- Amazon CloudWatch.
- AWS Auto Scaling to handle variable traffic loads.
- AWS Elastic Beanstalk

AI Image Processing Web Application

1. Scenario:

A startup company wants to launch a web-based application for their product catalogue. The application should be scalable, secure, and cost-efficient, while also ensuring reliable access to static and dynamic content. The architecture should support basic web hosting needs and be flexible enough to handle gradual user growth without heavy operational burden.

2. Problem Statement:

Design a simple, cost-effective, and scalable AWS-based architecture for hosting a web application that includes both static and dynamic content, provides secure access control, supports a relational database, and ensures good user experience under variable traffic loads.

3. Objectives:

- Host and serve static and dynamic content efficiently.
- Ensure secure access to AWS resources with proper permission controls.
- Enable scalability to handle variable traffic and growing user base.
- Use a relational database to store user and product data.
- Monitor system performance and ensure high availability.

Objective	Description
Static Content Hosting	Serve static files like images, HTML, and CSS using a reliable and low-cost service.
Backend Compute	Run backend logic on virtual machines with scalability and high availability.
Data Storage	Use a managed database to handle structured data.
Access Control	Enforce role-based access and resource protection.
Performance Optimization	Use Load Balancer and Auto Scaling for optimized traffic handling.
Monitoring	Track performance metrics and set up alerts using CloudWatch.

4. Outcomes:

- Static files (like images, HTML, CSS) are reliably stored and served from **Amazon S3**
- The backend application runs on **Amazon EC2** within a **VPC**, with traffic load balanced using **Elastic Load Balancer**.
- Access control and security are managed with **IAM** policies and roles.
- Relational data is stored securely using **Amazon RDS** within the VPC.
- Application can scale based on user demand using **AWS Auto Scaling**.
- System performance and availability are monitored with **Amazon CloudWatch**.

5. AWS Services Used:

Basic AWS Services:

- **Amazon S3:** To store and serve static content (e.g., images, JS, CSS, documents).
- **Amazon EC2:** For running the main application logic (backend services).
- **AWS IAM:** To manage user roles and restrict access (e.g., access to EC2, S3).
- **Amazon VPC:** To create a secure, isolated network environment.
- **Amazon RDS:** To manage the relational database (MySQL or PostgreSQL).

Advanced AWS Services:

- **Elastic Load Balancer (ELB):** To distribute incoming traffic across multiple EC2 instances for high availability.
- **AWS Auto Scaling:** To automatically increase or decrease the number of EC2 instances based on traffic.

6. Solution Architecture Design:

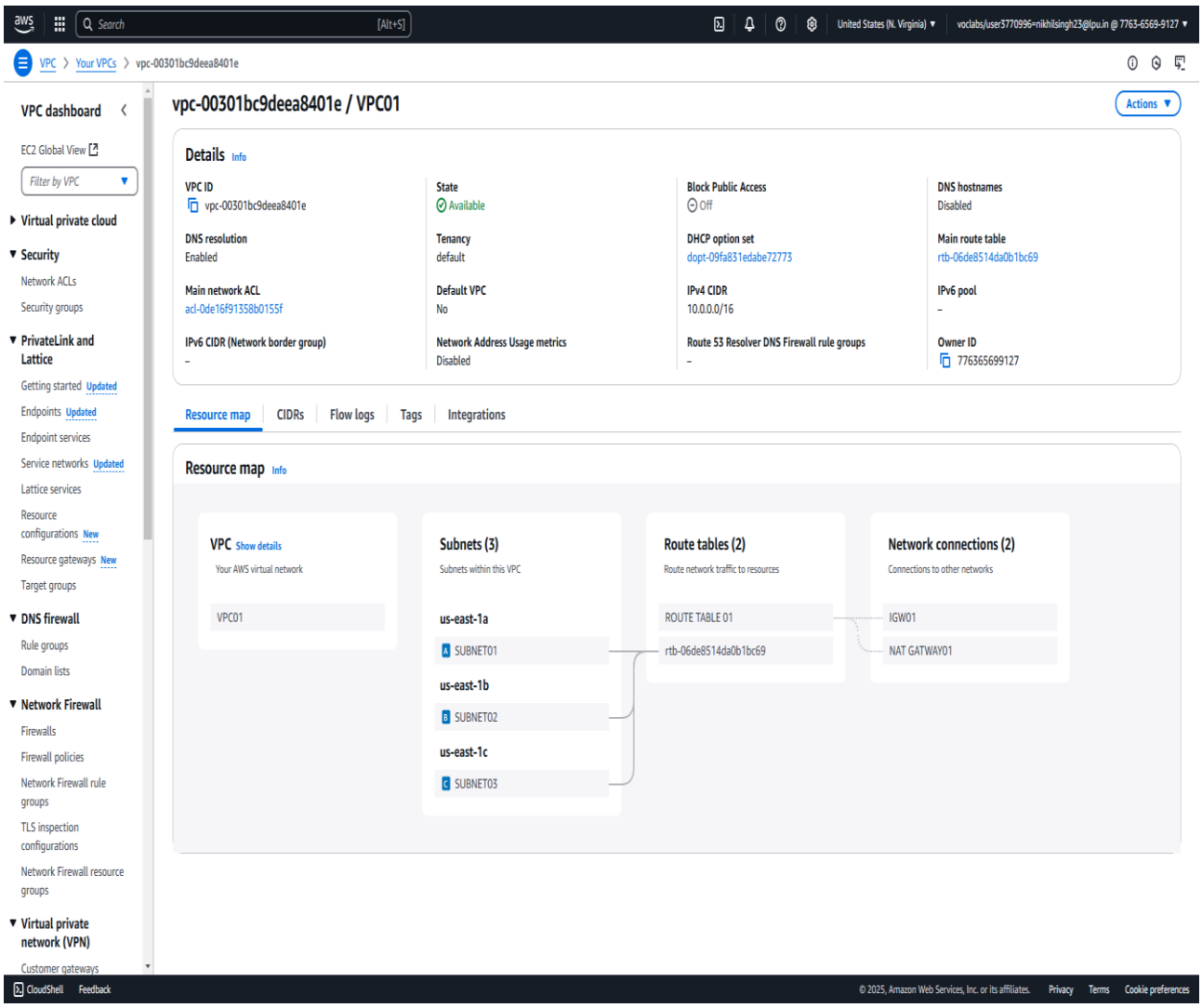
Implementation Steps:

Service-by-Service Breakdown

1. Amazon VPC (Virtual Private Cloud) – *Network Layer*

Purpose: Provides an isolated network environment where all services run securely.

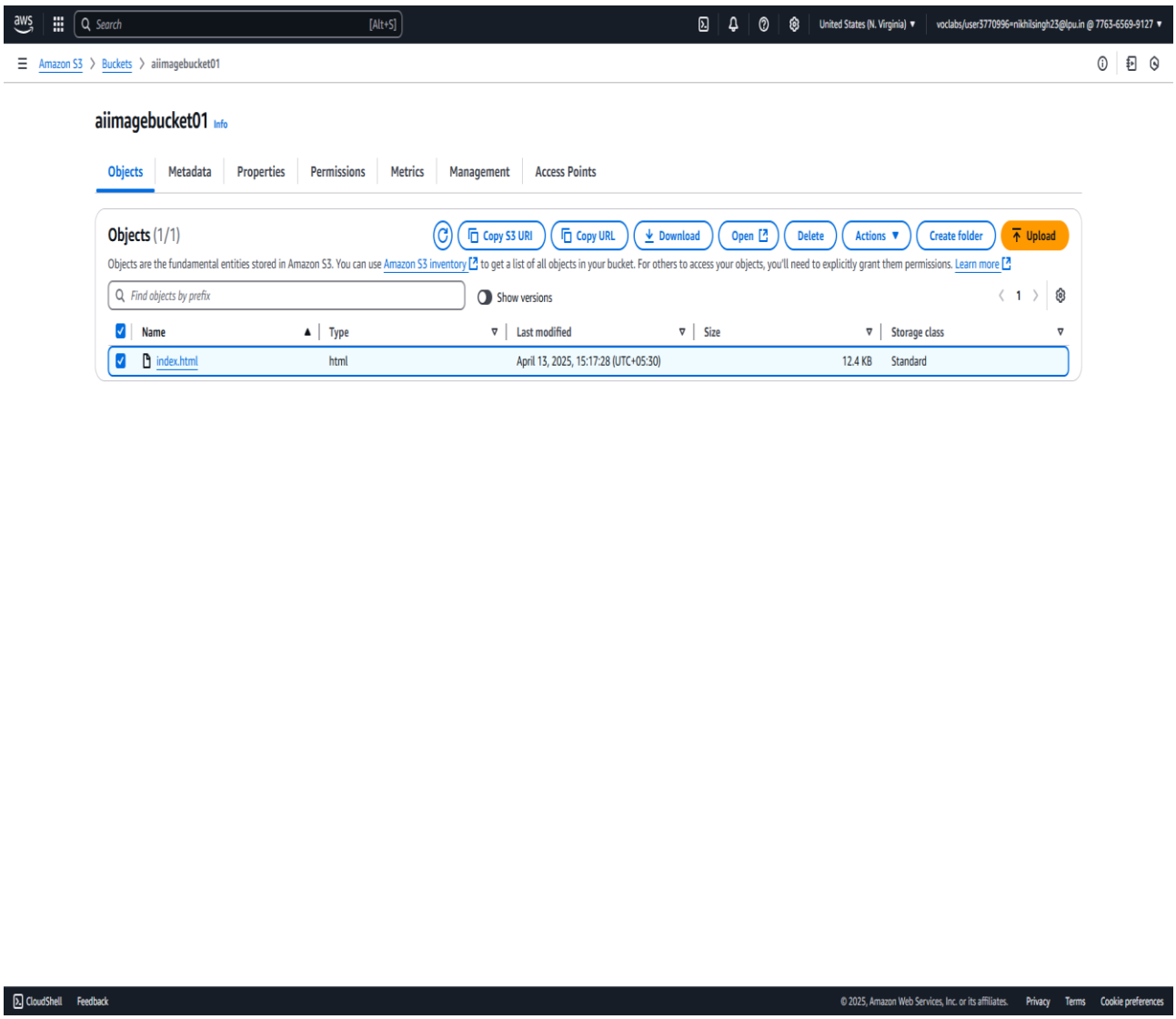
- **Components:**
  - 1 VPC (custom-created)
  - 2 Public Subnets (for EC2, Load Balancer)
  - 1 Private Subnet (for RDS)
  - Internet Gateway (for outbound traffic from public subnets)
  - NAT Gateway (optional for private subnet internet access)
  - Route Tables (for correct routing between public and private subnets)



2. Amazon S3 (Simple Storage Service) – *Static Content Storage*

Purpose: Store and serve static website files like HTML, images, JavaScript, and CSS.

- **Setup:**
  - Create an S3 Bucket (e.g., my app-static-files)
  - Enable Static Website Hosting
  - Upload all static files
  - Set bucket policy to allow public read access or restrict using CloudFront (optional)
  - Assign IAM Role to EC2 if it needs to read from S3 programmatically



3. Amazon EC2 (Elastic Compute Cloud) – *Application Layer*

Purpose: Host dynamic backend (e.g., Node.js, PHP, or Python app)

- **Setup:**
  - Launch EC2 instances in public subnet
  - Install and configure app server (e.g., Apache, Node.js)
  - Use Amazon Machine Image (AMI) for pre-configured setup
  - Assign IAM Role with permissions to access S3, CloudWatch, etc.
  - Ensure Security Group allows:
    - Port 80/443 (HTTP/HTTPS) from ELB
    - Port 22 (SSH) from your IP only

aws | Search [Alt+S] | United States (N. Virginia) | voclabs/user3770996=nikhilsingh23@lpu.in @ 7763-6569-9127

EC2 > Instances

To exit full screen, press and hold Esc

EC2

- Dashboard
- EC2 Global View
- Events
- **Instances**
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
- **Images**
  - AMIs
  - AMI Catalog
- **Elastic Block Store**
  - Volumes
  - Snapshots
  - Lifecycle Manager
- **Network & Security**
  - Security Groups
  - Elastic IPs
  - Placement Groups
  - Key Pairs
  - Network Interfaces
- **Load Balancing**
  - Load Balancers
  - Target Groups
  - Trust Stores
- **Auto Scaling**

Instances (1/2) Info

Last updated 1 minute ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
WEBSERVER01	i-0799b3b8e89cd03ac	Running	t2.micro	Initializing	View alarms +	us-east-1b	ec2-44-201-141-162.co...	44.201.141.162	–
Bastion Host	i-08dd254a38e6f9151	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-44-195-91-41.com...	44.195.91.41	–

i-0799b3b8e89cd03ac (WEBSERVER01)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID
i-0799b3b8e89cd03ac

IPv6 address
–

Hostname type
IP name: ip-172-31-93-52.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
44.201.141.162 [Public IP]

Public IPv4 address
44.201.141.162 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-93-52.ec2.internal

Instance type
t2.micro

VPC ID
vpc-03e0d4ad297be8ba6

Private IPv4 addresses
172.31.93.52

Public IPv4 DNS
ec2-44-201-141-162.compute-1.amazonaws.com | open address

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

aws | Search [Alt+S] | United States (N. Virginia) | voclabs/user3770996=nikhilsingh23@lpu.in @ 7763-6569-9127

```
       __|  __|_  )
       _|  (     /   Amazon Linux 2023
      ___|\___|___|
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-93-52 ~]$
```

i-0799b3b8e89cd03ac (WEBSERVER01)

PublicIPs: 44.201.141.162 PrivateIPs: 172.31.93.52

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4. Elastic Load Balancer (ELB) – *Traffic Distribution*

Purpose: Distribute incoming traffic across multiple EC2 instances.

- **Setup:**
  - Create Application Load Balancer
  - Register EC2 instances with the target group
  - Configure health checks (e.g., /health)
  - Attach Load Balancer to Auto Scaling Group
  - Allow traffic from Load Balancer in EC2's Security Group

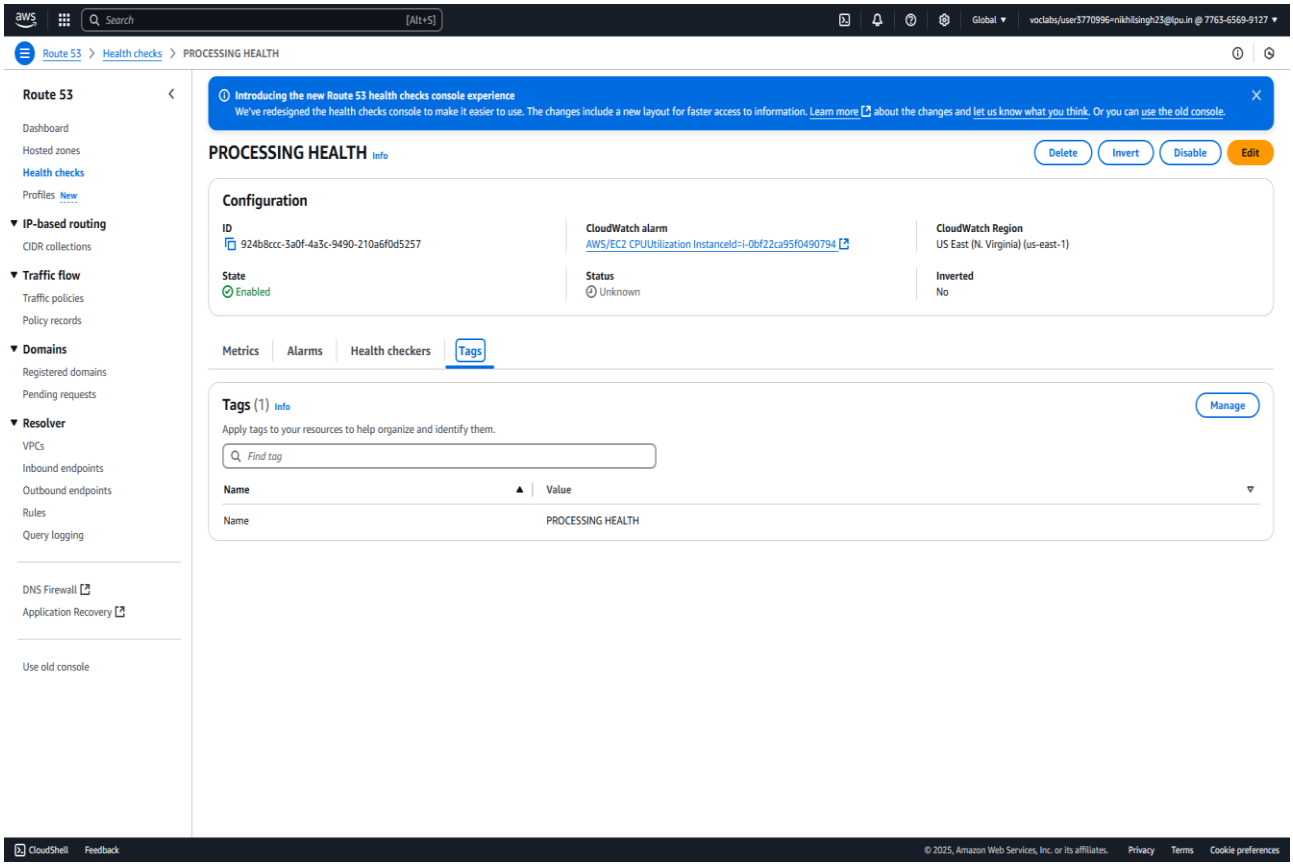
Route 53 (with EC2 Health Checks and DNS-Based Load Balancing)

Use Case: Simple load balancing across multiple EC2 instances.

How it works:

- You create multiple EC2 instances in different availability zones.
- Use Route 53 to configure a DNS "weighted" or "failover" routing policy.
- Health checks ensure only healthy instances receive traffic.

Best for: Simple, low-traffic apps needing failover capabilities.



5. AWS Auto Scaling – *Automatic Scalability*

Purpose: Automatically scale the number of EC2 instances based on traffic/load.

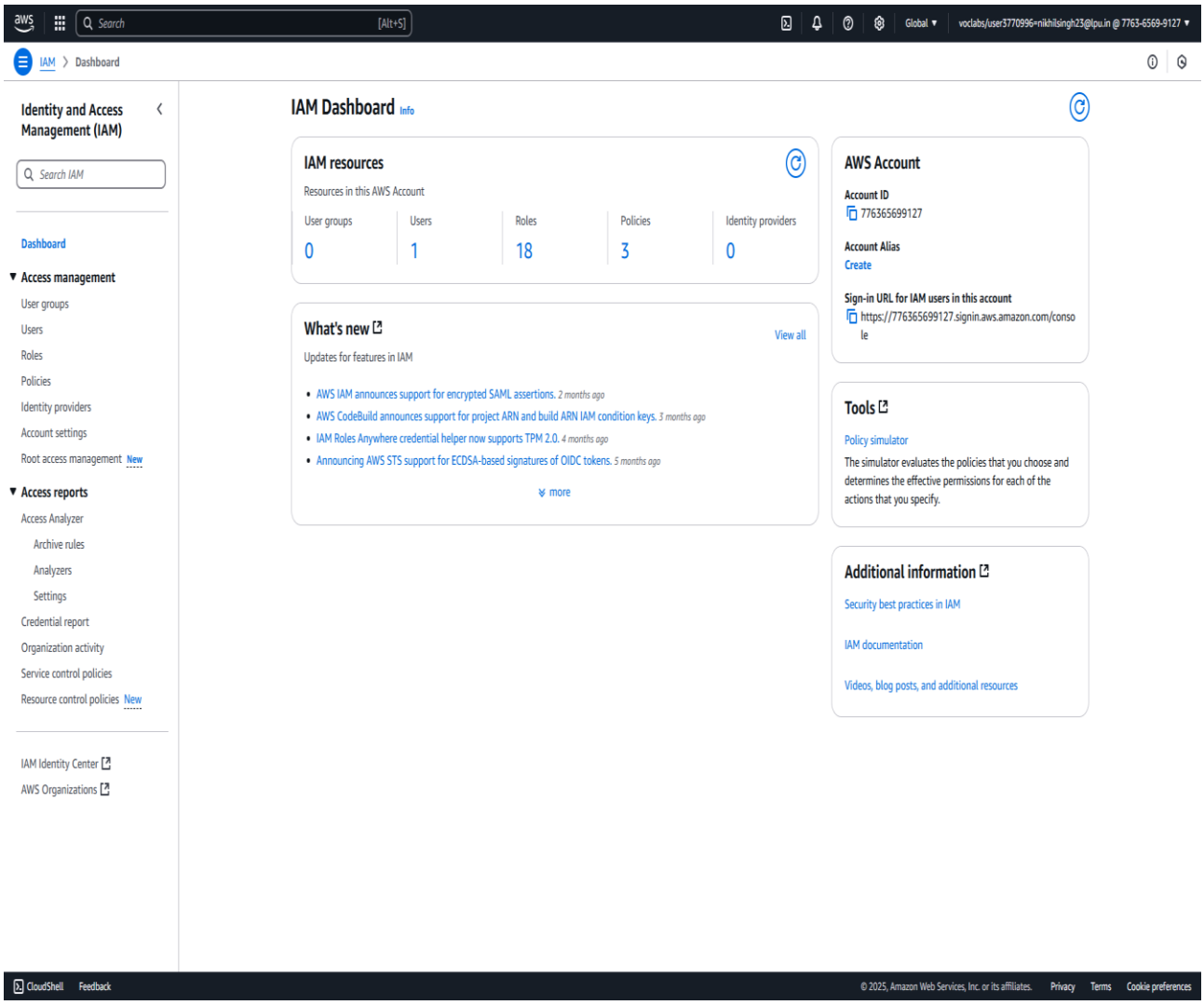
- **Setup:**
  - Create an Auto Scaling Group linked to your ELB
  - Define launch template for EC2 instance
  - Configure scaling policy:
    - e.g., Add 1 instance if CPU > 70%
    - Remove 1 instance if CPU < 20%
  - Set minimum and maximum instance limits (e.g., Min: 2, Max: 6

6. AWS IAM (Identity and Access Management) – *Security Control*

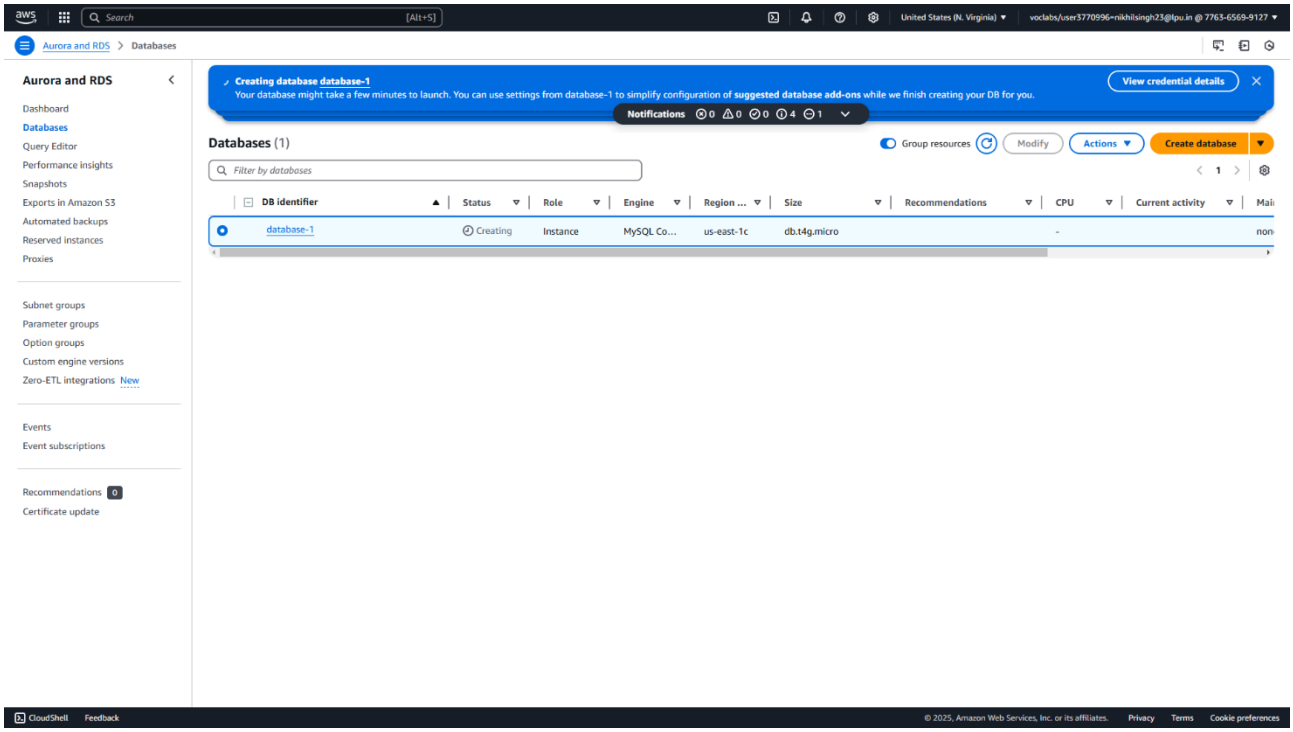
Purpose: Control who can access AWS services and resources securely.

- **Setup:**
  - Create IAM users with least privilege policies
  - Create IAM Roles for EC2 and Lambda with required permissions
  - Use IAM Policies to restrict S3, RDS, EC2, etc.

- Enable MFA (Multi-Factor Authentication) for sensitive users



RDS:



7. Amazon RDS (Relational Database Service) – *Data Storage*

Purpose: Host a relational database securely inside the VPC.

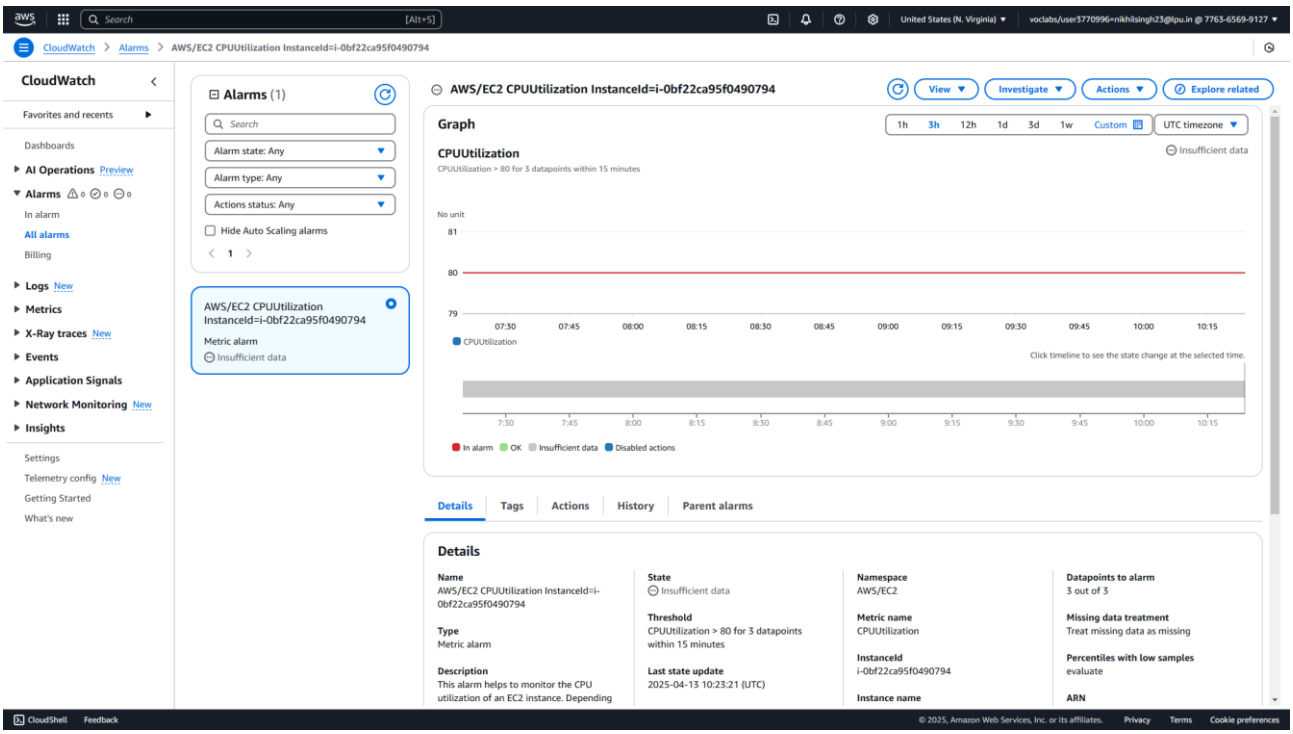
- **Setup:**
    - Launch MySQL or PostgreSQL RDS in a private subnet
    - Enable multi-AZ deployment for high availability
    - Enable Automated Backups

    - Allow only EC2 security group to access the RDS endpoint
    - Optionally enable encryption at rest

8. Amazon CloudWatch – *Monitoring and Logging*

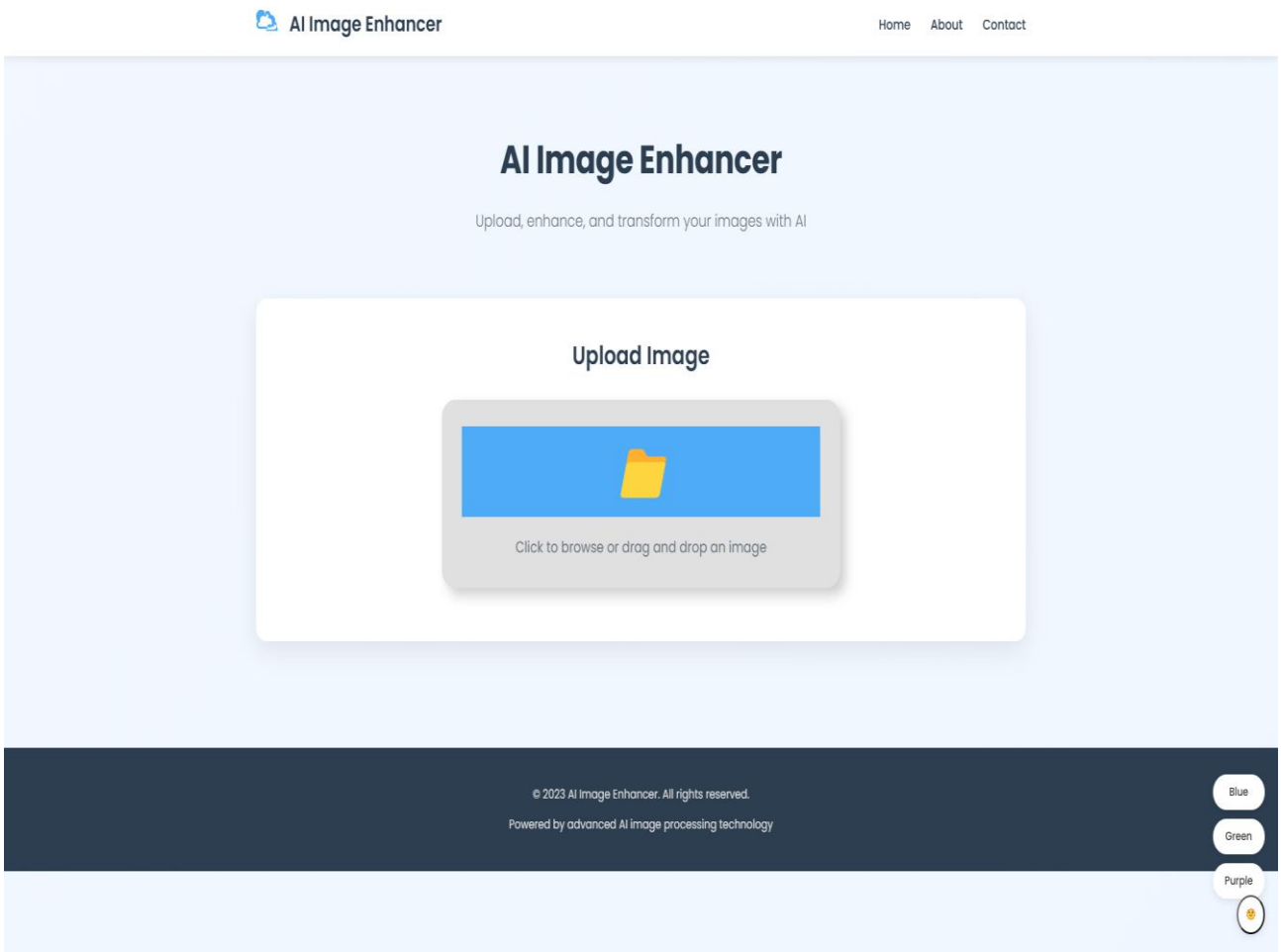
Purpose: Track metrics, logs, and set alarms for your system.

- **Setup:**
    - Enable detailed monitoring on EC2 and RDS
    - Set up CloudWatch Alarms for:
        - High CPU Usage
        - RDS storage thresholds
        - Auto Scaling events



- Use CloudWatch Logs Agent on EC2 to send logs (e.g., Apache/Nginx logs)

Webpage:



9. Architecture Diagram Summary:

```
+-------------------------+
|      End Users          |
+-----------+-------------+
            |
            v
+------------------Elastic Load Balancer-------------------+
|                                                           |
+------v------+                          +----------v--------+
|  EC2 App   |                           |    EC2 App        |
| Server 1   |  Auto Scaling Group (min: 2)      |   Server 2        |
+------+------++-----------------------------------++------------------+
|                                                           |
+---------------------------+-------------------------+
                            |
                    +-------v--------+
                    |    Amazon RDS |
                    | (MySQL)|   <-- (Private Subnet)
                    +-----------------+
                            |
                    +-----v-----+
                    |   VPC     |
```

```
+-----+-----+
|
+---------v---------+
| Amazon S3 | <-- (Static Files)
+------------------+
|
+----------v------------+
| IAM & Security Groups |
+------------------------+
|
+-------v-------+
| CloudWatch |
+---------------+
```

10. Benefits of This Architecture:

- **Scalable:** Auto Scaling handles traffic spikes automatically.
- **Highly Available:** ELB distributes traffic and RDS multi-AZ ensures DB reliability.
- **Cost-Effective:** Only pay for what you use; S3 for cheap storage, Auto Scaling for optimized compute costs.
- **Secure:** IAM, VPC, and security groups ensure proper access control.
- **Easy to Extend:** Future services like Lambda, CloudFront, or Beanstalk can be easily integrated.

11. Conclusion:

This architecture provides a **fully functional, secure, and scalable AWS environment** using both basic and advanced AWS services:

Feature	AWS Services Used
Static File Hosting	Amazon S3
Backend Compute	Amazon EC2
Load Balancing	Elastic Load Balancer
Auto Scaling	AWS Auto Scaling
Database	Amazon RDS
Network Setup	AWS VPC
Access Control	AWS IAM
Monitoring	Amazon CloudWatch

THANKYOU!